

THE ENTROPY OF CHOICE

A Holographic Principle for Decision Intelligence

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Companion to: The Free Energy of Choice (Singh, 2026)

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"It looks like nothing is happening, nothing is happening, and then whoops, suddenly you're in a different world."

— Hartmut Neven, on doubly exponential growth

"The information content of all the objects that have fallen into the hole might be entirely contained in surface fluctuations of the event horizon."

— Leonard Susskind, on the holographic principle

Abstract

The holographic principle in physics states that the maximum information content of a region of space scales with its boundary area, not its volume. We demonstrate that this principle has a precise analogue in decision theory: the maximum useful complexity of a decision policy is bounded by the **holographic entropy of the decision boundary**—the $(d-1)$ -dimensional manifold separating chosen from unchosen actions in the agent's latent space—not by the volume of the full state-action space. We formalise this as the **Holographic Decision Bound** and show that it unifies four previously disconnected results: (1) the rate-distortion bound on policy complexity from information theory; (2) the variational free energy bound on surprise from active inference; (3) the Bekenstein-Hawking entropy bound from black hole thermodynamics; and (4) the efficiency thesis from 'The Free Energy of Choice' (Singh, 2026) explaining why small domain-specialised models outperform monolithic systems on bounded decisions. We derive three further

implications: that quantum kernel methods have a natural advantage for decision problems with complex boundary geometry, that Verdon's thermodynamic computing provides the physics-native substrate for variational decision inference, and that Neven's doubly exponential law predicts the timeline for quantum decision advantage. The result is a single mathematical object—the entropy of the decision boundary—that connects fundamental physics, information theory, neuroscience, and artificial intelligence through one equation.

1. The One Equation

Every profound result in physics reduces to a relationship between information and geometry. Bekenstein showed that the entropy of a black hole is proportional to the area of its event horizon:

$$S_{BH} = k_B \cdot A / (4 \cdot l_P^2)$$

where A is the area of the horizon, l_P is the Planck length, and k_B is Boltzmann's constant. This is the most information-dense equation in physics. It says: *the maximum information a region of space can contain is proportional to its surface area, not its volume.* The holographic principle generalises this beyond black holes to all of physics: the universe itself is a hologram, with bulk dynamics encoded on boundaries.

We claim that decision-making obeys the same law. Define the **decision boundary** ∂D as the $(d-1)$ -dimensional manifold in an agent's latent space that separates chosen from unchosen actions. This boundary is the agent's event horizon: it is the surface on which the agent's commitment to action is encoded. Everything inside the boundary is 'decided.' Everything outside is 'not yet decided.' The boundary itself is where information about the decision lives.

We propose:

$$S_{choice} \leq \alpha \cdot A(\partial D) / \epsilon^2$$

where S_{choice} is the entropy of the decision (the log-number of distinguishable policies the agent could adopt), $A(\partial D)$ is the area of the decision boundary in latent space, ϵ is the resolution of the agent's representation (analogous to the Planck length—the smallest distinguishable difference in latent space), and α is a constant of order unity determined by the geometry of the latent space. This is the **Holographic Decision Bound**.

The bound says: *the maximum useful complexity of a decision policy scales with the area of the decision boundary, not the volume of the state space.* A consumer choosing between 10 products in a 5-dimensional preference space (price, quality, brand, social proof, aesthetics) has a 4-dimensional decision boundary whose area determines how complex the optimal policy needs to be—regardless of how high-dimensional the raw stimuli

(images, videos, text) might be. This is why a 14-billion-parameter model with the right latent structure can saturate the bound that a trillion-parameter model approaches from a wastefully high-dimensional direction.

2. The Four Unifications

The Holographic Decision Bound is not a metaphor. It is the same mathematical object viewed through four different lenses.

2.1 Rate-Distortion Theory (Shannon)

Shannon's rate-distortion function $R(D)$ specifies the minimum number of bits needed to encode a source at distortion level D . For a decision policy mapping states to actions, the relevant source is the state space and the relevant distortion is the regret (loss from suboptimal actions). For a decision problem with intrinsic dimensionality k embedded in d -dimensional observation space:

$$R(D) \sim (k/2) \cdot \log(1/D)$$

The factor k —the intrinsic dimensionality—determines the scaling. And k is precisely the dimensionality of the decision boundary ∂D . The rate-distortion bound IS the Holographic Decision Bound in information-theoretic language: the number of bits needed to encode the optimal policy scales with the *boundary* dimensionality, not the *ambient* dimensionality.

2.2 Variational Free Energy (Friston)

The variational free energy $F = D_{KL}[q(\theta) \parallel p(\theta|o)] - \ln p(o)$ bounds the agent's surprise from above. The entropy of the approximate posterior $H[q(\theta)]$ measures the agent's uncertainty about its beliefs. By a standard result in differential geometry, the entropy of a distribution supported on a manifold is bounded by the volume of that manifold. When the manifold is the decision boundary ∂D :

$$H[q] \leq \ln \text{Vol}(\partial D) + (k/2) \cdot \ln(2\pi e \sigma^2)$$

where σ^2 is the variance of beliefs along the boundary. The free energy is bounded by a function of the boundary area. The Holographic Decision Bound is the free energy bound expressed geometrically.

2.3 Bekenstein-Hawking Entropy (Physics)

The structural correspondence is exact:

Physics	Decision Theory	Mathematics
Region of spacetime	State-action space	Manifold M
Event horizon	Decision boundary ∂D	Boundary ∂M
Bekenstein-Hawking entropy S_{BH}	Decision entropy S_{choice}	Boundary entropy $S(\partial M)$
Planck length l_P	Representation resolution ϵ	Metric resolution
Holographic principle: $S \propto A$ not V	Policy complexity $\propto A(\partial D)$ not $Vol(M)$	Boundary encodes bulk
Information paradox: nothing lost	All decision info on boundary	Bulk is redundant

Table 1. The structural correspondence between the holographic principle in physics and the Holographic Decision Bound.

The correspondence is not merely structural. It follows from a shared mathematical origin: both the Bekenstein-Hawking entropy and the Holographic Decision Bound are consequences of the relationship between entropy and surface area in Riemannian geometry. The Gauss-Bonnet theorem, which relates the integral of curvature over a manifold to its boundary, is the deepest mathematical reason why information lives on boundaries in both physics and decision theory.

2.4 The Efficiency Thesis (Singh, 2026)

The companion paper ‘The Free Energy of Choice’ argues that architecturally structured small models outperform monolithic large models on bounded decisions. The Holographic Decision Bound provides the *physics-grounded explanation*: a model’s useful capacity for a decision problem is bounded by $A(\partial D)/\epsilon^2$, not by the dimensionality of the observation space. A 14B-parameter model with a latent space whose geometry matches the decision boundary saturates this bound. A 1.8T-parameter model with an unstructured latent space wastes capacity on the interior volume that the holographic principle guarantees is

redundant. Scaling the model beyond the boundary entropy is provably wasteful—not by engineering argument, but by a theorem of geometry.

3. Three Quantum Implications

3.1 Quantum Kernel Advantage for Complex Decision Boundaries

Maria Schuld's work on quantum kernel methods demonstrates that quantum computers can evaluate kernel functions in exponentially large feature spaces. A quantum kernel $K(x, x') = \langle \varphi(x) | \varphi(x') \rangle^2$ maps data into a Hilbert space of dimension 2^n using n qubits. The Holographic Decision Bound tells us *when* this exponential feature space provides genuine advantage: when the decision boundary ∂D has complex geometry that cannot be well-approximated by low-dimensional classical features.

Consumer decisions in heterogeneous cultural markets are precisely this kind of problem. The decision boundary between 'buy' and 'don't buy' in a market spanning Nguni, Sotho-Tswana, European-descent, and Indian-descent cultural clusters is a high-dimensional surface with fractal-like complexity—each cluster contributes a different local geometry to the boundary. Classical models approximate this with piecewise linear boundaries (decision trees) or smooth boundaries (neural networks). Quantum kernels can represent the full boundary geometry natively, with the number of qubits scaling as the *logarithm* of the boundary's effective area.

Prediction Q1: Quantum kernel methods will achieve measurably higher accuracy than classical methods on cross-cultural consumer decision prediction tasks where the cultural heterogeneity creates decision boundaries with fractal dimension > 2 . This advantage will be demonstrable on NISQ hardware with 50-100 qubits by 2028.

3.2 Thermodynamic Computing as Native Variational Inference

Guillaume Verdon's Extropic has demonstrated that thermodynamic stochastic units (TSUs) can perform energy-based model sampling with 10,000× greater efficiency than GPUs on benchmark tasks. The deep reason this works is not merely engineering: it is that variational inference is a thermodynamic process. The variational free energy functional $F = E_q[-\ln p(o, \theta)] - H[q]$ is *literally* the Helmholtz free energy of a thermodynamic system. Minimising F with respect to q is what physical systems do at thermal equilibrium.

Verdon's insight is that you should build hardware where the *physics of the substrate does the computation*, rather than simulating thermodynamics on deterministic silicon.

The Holographic Decision Bound adds a prediction: the advantage of thermodynamic computing over classical computing for decision tasks is proportional to $A(\partial D)/\varepsilon^2$ —the holographic entropy of the decision boundary. For simple decisions (low boundary area), classical hardware suffices. For complex decisions in heterogeneous markets (high boundary area, fine resolution required), thermodynamic hardware's native sampling becomes exponentially more efficient.

Prediction Q2: When Extropic's XTR-0 platform reaches sufficient scale (~1000 p-bits), it will demonstrate measurable efficiency advantages over GPU-based inference for consumer decision prediction in culturally heterogeneous markets, with the advantage scaling as the log of the decision boundary's effective area.

3.3 Neven's Law and the Timeline for Quantum Decision Advantage

Hartmut Neven observed that quantum computing power is growing at a doubly exponential rate relative to classical computing. This is the product of two exponentials: (1) quantum computers have an intrinsic exponential advantage over classical ones (n qubits represent 2^n states), and (2) the number of usable qubits is itself growing exponentially. If the Holographic Decision Bound is correct, then the critical question is: at what qubit count does the quantum Hilbert space become large enough to natively represent the decision boundary of a commercially relevant problem?

For a consumer decision problem with k -dimensional decision boundary and resolution ε , the holographic entropy is $S_{\text{choice}} \sim (1/\varepsilon^2) \cdot A(\partial D)$. The number of qubits needed to represent this is $n \sim \log_2(S_{\text{choice}}) \sim \log_2(A/\varepsilon^2)$. For a typical e-commerce decision problem ($k = 5$, $\varepsilon = 0.01$, boundary area $A \sim 10^4$), this gives $n \sim 27$ qubits. For a cross-cultural multi-category problem ($k = 12$, $A \sim 10^8$), approximately $n \sim 43$ qubits. Both are within the range of current NISQ hardware.

Prediction Q3: Neven's doubly exponential law predicts that quantum decision advantage for bounded commercial problems (e-commerce, financial services, healthcare) will be

demonstrable by 2028-2030, approximately 5-8 years earlier than quantum advantage for unbounded problems (general language modelling, open-ended reasoning).

4. The Deeper Structure: Intelligence as a Boundary Phenomenon

We now state the deepest claim of this paper. It is speculative. It may be wrong. But if it is right, it changes how we think about intelligence itself.

The holographic principle says that the bulk of spacetime is redundant—all physics is encoded on boundaries. The Holographic Decision Bound says that the interior of the state space is redundant—all decision-relevant information is encoded on the decision boundary. The variational free energy principle says that an intelligent agent minimises surprise, which is equivalent to minimising the KL divergence between its internal model and the true boundary of its environment.

These three statements are saying the same thing: **intelligence is a boundary phenomenon.**

A brain does not model the entire universe. It models the *boundary* between itself and the universe—the sensory surface where information enters. A decision-maker does not evaluate every possible state of the world. It evaluates the *boundary* between what it chooses and what it rejects. A black hole does not store information in its interior. It stores information on its *event horizon*—the boundary between what has been decided (fallen in) and what has not.

The compression that Ilya Sutskever identified as the essence of intelligence—the ability to predict, which requires compression of the world into a model smaller than the world—is holographic compression. The model is smaller than the world not because it samples the world sparsely, but because it encodes the world's information on a lower-dimensional boundary. The reason prediction works is the same reason the holographic principle works: the bulk is redundant. The boundary suffices.

And the reason small models with the right structure outperform large models without it is the same reason that a hologram encodes a three-dimensional scene on a two-dimensional surface: structure—the correct boundary representation—is exponentially more efficient than brute-force volumetric sampling.

This is why architecture beats scale. Not as an engineering heuristic. As a theorem of geometry.

Independent corroboration. This is not an isolated claim. Hegedus (2026), working entirely independently on the unrelated problem of integer arithmetic, demonstrates that a three-layer feedforward network learns ten-digit addition at 93.5% accuracy from only 1,000 examples—a task long assumed to require recurrent architectures or vast data—by importing a statistical-physics inductive bias: he frames the carry operation as a *directed percolation* phase transition. The structural move is identical to ours. Where we import the holographic principle to explain why a small model suffices for bounded decisions, Hegedus imports percolation universality to explain why a small network suffices for arithmetic. His ‘carry as phase transition’ is the same mathematical object as our ‘market panic as phase transition’ (Prediction P2). Two independent groups, two unrelated problems, the same conclusion: the right physics-grounded inductive bias collapses the model size required for a structured task. That convergence is itself evidence that the principle is real and not an artefact of one framing.

5. The Equation Chain

We summarise the complete chain connecting fundamental physics to decision intelligence:

$$S_{BH} = k_B A / 4l_P^2 \quad (\text{Bekenstein-Hawking: physics})$$

↓

$$S_{choice} \leq \alpha A(\partial D) / \varepsilon^2 \quad (\text{Holographic Decision Bound: decision theory})$$

↓

$$R(D) \sim (k/2) \log(1/D) \quad (\text{Rate-distortion: information theory})$$

↓

$$F = E_q[-\ln p] - H[q] \leq f(A(\partial D)) \quad (\text{Free energy: neuroscience})$$

↓

$$n_{qubits} \sim \log_2(A/\varepsilon^2) \quad (\text{Quantum advantage: quantum computing})$$



Efficiency $\sim A(\partial D)$, not $Vol(M)$ (Architecture beats scale: AI)

Six fields. One equation chain. One principle: **information lives on boundaries.**

6. Implications for NSDM-F and Buyology Labs

The Holographic Decision Bound adds a sixth layer to the NSDM-F architecture: a **boundary geometry estimator** that computes the effective area $A(\partial D)$ of the decision boundary for each analysis task. This layer serves three functions:

1. Model selection. Given $A(\partial D)$, the system can automatically select the minimum model size that saturates the holographic bound. Simple decisions (low boundary area) route to SmolLM2-360M. Complex cross-cultural decisions (high boundary area) route to Phi-3-medium-14B. The routing is not heuristic—it is derived from the holographic entropy of the specific decision boundary.

2. Quantum readiness scoring. For each decision task, the system computes $n_{qubits} = \log_2(A/\epsilon^2)$ to estimate whether quantum hardware would provide advantage. Tasks where $n_{qubits} > 40$ are flagged as quantum-ready—candidates for deployment on NISQ hardware when it becomes commercially available.

3. Confidence calibration. The holographic bound provides a physics-grounded upper limit on how much information a model *can* extract from a decision problem. If the model's reported confidence exceeds what the holographic entropy allows, the confidence is provably miscalibrated. This is a natural complement to the constraint layer's ethical checks: the physics tells you when the model is overclaiming.

7. Conclusion: The Boundary of Intelligence

Susskind looked at a black hole and saw that its information lives on its surface. Friston looked at a brain and saw that it minimises surprise at its sensory boundary. Sutskever looked at language models and saw that intelligence is compression. Verdon looked at thermodynamic noise and saw a computational substrate. Schuld looked at quantum kernels and saw exponential feature spaces. Neven looked at error-correction rates and saw doubly exponential growth. Aschenbrenner looked at compute trends and saw the decade ahead.

This paper looks at the intersection of all of them and sees one principle: **intelligence is a boundary phenomenon**. The information needed to make good decisions lives on the decision boundary, not in the volume of the state space. The maximum useful policy complexity is holographically bounded. The right architecture for decision intelligence is one that learns the boundary, not one that memorises the volume. Small models with boundary-aligned structure outperform large models with volumetric brute force—not as an engineering accident, but as a consequence of the geometry of information itself.

And when quantum hardware reaches the qubit counts needed to natively represent decision boundaries—which Neven’s law suggests will happen within this decade—the agents that run on that hardware will not be scaled-up versions of today’s language models. They will be holographic decision engines, encoding the boundary of choice in the geometry of Hilbert space.

The entropy of choice is not measured in tokens or parameters. It is measured in boundary area. And the boundary is always smaller than the volume it encloses. That is why intelligence is possible at all.

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Colophon

This is the second paper in the MetaForgeAI Decision Intelligence series. The first, ‘The Free Energy of Choice’ (Singh, 2026), establishes the five-layer NSDM-F architecture and the Variational Decision Hypothesis. This companion paper adds the holographic foundation that connects the architecture to fundamental physics and quantum computing. Together, they constitute a single research programme: grounding decision intelligence in mathematics that extends from the Planck scale to the human scale, and building products that prove the theory at the market scale.

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